

NAVIN BHATTARAI

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RESEARCH INTERESTS

Thermal-Fluid Systems, Heat Transfer, Flow-Channel Optimization, Thermal Management, Computational Fluid Dynamics (CFD), Topology Optimization, Computational Mechanics, Numerical Methods

EDUCATION

B.S. in Mechanical Engineering

Expected Dec. 2026

Washington University in St. Louis, Missouri

GPA: 3.98/4.00

- Honors: *Lee Hunter Scholarship*
- Relevant Coursework: Heat Transfer, Computational Fluid Dynamics, Thermo-Fluid Modeling (Renewable Energy Systems), Design of Thermal Systems, Fluid Mechanics, Thermodynamics, Control Systems, Solid Mechanics, Machine Elements

B.A. in Physics

Degree Awarded May 2026

Truman State University, Kirksville, Missouri

GPA: 4.00/4.00

- Completed via the Truman State–Washington University 3+2 Dual-Degree Engineering Program (resident at Truman State: August 2022 – May 2025)
- Honors: *Summa Cum Laude, Sigma Pi Sigma, International President's Honorary Scholarship, 7+ Merit-Based Scholarships*
- Advanced coursework in mathematical methods, classical mechanics, thermodynamics, and experimental physics

RELEVANT RESEARCH EXPERIENCE

Undergraduate Researcher, SURGE Program

May 2026 – Present

Washington University in St. Louis

St. Louis, MO

Advisor: Dr. Xianglin Li

Research Project: Flow Channel Optimization for Heat and Mass Transfer

- Conducting computational fluid dynamics research on flow-channel optimization for enhanced heat and mass transfer
- Developing OpenFOAM workflows spanning geometry creation, mesh generation, solver configuration, convergence monitoring, and post-processing using ParaView
- Implementing topology optimization using the **Method of Moving Asymptotes (MMA)** to generate optimized flow-channel architectures
- Evaluating designs using pressure drop, temperature distribution, heat-transfer performance, and thermal uniformity
- Validating computational models against published benchmark studies

Research Assistant

Feb. 2024 – Aug. 2025

Truman State University, Department of Physics

Kirksville, MO

Advisor: Dr. Rasanjali Jayathissa

Research Project: Dye-Sensitized Solar Cells Using Natural-Dye Sensitizers

- Investigated plant-derived natural dyes as alternative sensitizers for dye-sensitized solar cells (DSSCs)
- Designed experiments for natural-dye extraction and characterization using UV-Visible spectroscopy and Fourier Transform Infrared Spectroscopy (FTIR)
- Fabricated and tested **30+** laboratory-scale DSSCs and characterized performance using Electrochemical Impedance Spectroscopy (EIS) and current-voltage measurements from a solar simulator
- Analyzed experimental data to evaluate relationships among dye properties, material characteristics, and photovoltaic performance
- Presented findings at the 2025 APS Prairie Section Meeting

RELEVANT PRESENTATIONS

- **Bhattarai, Navin.** “Harnessing the power of natural dyes for clean energy: Dye-Sensitized Solar Cells based on plant dyes.” *APS Prairie Section Meeting*, 2025.

RELEVANT TEACHING EXPERIENCE

Undergraduate Course Assistant

Aug. 2025 – Present

McKelvey School of Engineering, Washington University in St. Louis

St. Louis, MO

- Support undergraduate instruction in Statics, Mechanics of Materials, Solid Mechanics, and Thermodynamics across classes of approximately **80–150 students**
- Conduct weekly office hours, assist students with problem-solving, and provide individualized technical guidance on course material
- Grade homework, assignments, and exams; proctor examinations and assist instructors with course administration and assessment
- Currently supporting Thermodynamics and Solid Mechanics; previously supported Statics and Mechanics of Materials

Academic Peer Tutor

Aug. 2023 – May 2025

Center of Academic Excellence, Truman State University

Kirkville, MO

- Tutored Calculus I–III and introductory Physics I–II through individual and small-group sessions across multiple semesters
- Provided in-class embedded tutoring during Fall 2024, supporting students during class activities while reviewing and reinforcing course material
- Adapted tutoring strategies to different courses and student needs, emphasizing conceptual understanding and problem-solving approaches
- Collaborated with faculty and instructional staff to align tutoring support with course objectives

ENGINEERING & PROFESSIONAL EXPERIENCE

Co-Lead

Aug. 2025 – Present

WashU Design/Build/Fly, Structures Sub-Team

St. Louis, MO

- Design and model aircraft structural components and subassemblies using SOLIDWORKS for fabrication and integration
- Optimize structural designs for manufacturability and collaborate with aerodynamics and propulsion teams to satisfy performance requirements

Level 1 Technician

Feb. 2026 – Present

Student Technology Services, Washington University in St. Louis

St. Louis, MO

- Diagnose and resolve hardware, software, network, and account-access issues using systematic troubleshooting

Building Operations Manager

Aug. 2022 – Aug. 2025

Student Union and Involvement Services, Truman State University

Kirkville, MO

- Coordinated **100+ events** for **45+ organizations** while managing equipment setup, facility logistics, safety requirements, and operations
- Trained and supervised **6+ assistants** while enforcing facility standards and operational procedures

HONORS & AWARDS

- Summa Cum Laude, Truman State University, 2026
- Lee Hunter Scholarship, Washington University in St. Louis
- Sigma Pi Sigma, Physics Honor Society
- International President’s Honorary Scholarship
- 7+ Merit-Based Academic Scholarships

SKILLS

CFD & Computational: OpenFOAM, ParaView, Computational Fluid Dynamics, Topology Optimization, Method of Moving Asymptotes (MMA)

Programming: Python, C++, MATLAB, Mathematica, R

Mechanical Design: SOLIDWORKS, AutoCAD, Onshape, Finite Element Analysis (FEA), Technical Drawing, DFM/DFA, Tolerance Analysis

Experimental Methods: UV-Vis Spectroscopy, FTIR, Electrochemical Impedance Spectroscopy, Solar Simulator I-V Testing, Prototype Fabrication, Data Analysis

Manufacturing: Machining, FDM 3D Printing, Soldering